

OSSP-3

Question:

3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations

Solution:

```
GNU nano 8.7.1                                practical3.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid < 0)
    {
        printf("Fork Failed\n");
        return 1;
    }

    if(pid == 0)
    {
        printf("\n----- CHILD PROCESS ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child Running...\n");

        sleep(10);

        printf("Child Terminated\n");
    }
    else
    {
        printf("\n----- PARENT PROCESS ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Parent PPID : %d\n", getppid());

        wait(NULL);

        printf("Parent Resumed After Child Completion\n");
    }

    return 0;
}

ubuntu@ubuntu:~$ nano practical3.c
ubuntu@ubuntu:~$ gcc practical3.c -o practical3
ubuntu@ubuntu:~$ ./practical3

----- PARENT PROCESS -----
Parent PID : 7590
Parent PPID : 5772

----- CHILD PROCESS -----
Child PID : 7591
Child PPID : 7590
Child Running...
Child Terminated
Parent Resumed After Child Completion
```

```
ubuntu@ubuntu:~$ ps -ef | grep practical3
ubuntu 7615 5772 99 08:25 pts/0 00:00:00 grep --color=auto practical3
ubuntu@ubuntu:~$ ps -o pid,ppid,state,cmd -p 2301,2302
PID PPID S CMD
ubuntu@ubuntu:~$ top

top - 08:26:54 up 3:20, 1 user, load average: 0.27, 0.09, 0.02
Tasks: 230 total, 1 running, 229 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.3 us, 0.7 sy, 0.0 ni, 97.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3892.8 total, 81.3 free, 2767.3 used, 1499.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1125.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM     TIME+ COMMAND
 3356 ubuntu    20   0 5175288 488136 165360 S   6.3  12.2   1:45.42 gnome-shell
 5788 ubuntu    20   0 2930804 492768 137376 S   5.3  12.4   1:06.71 ptxcli
 4651 root       20   0 94444 8936 5784 S   0.3   0.2   0:01.15 spice-vdagentd
 7625 ubuntu    20   0 24028 7244 3696 R   0.3   0.2   0:01.54 top
    1 root      20   0 26656 17064 10552 S   0.0   0.4   0:04.34 systemd
    2 root      20   0 0 0 0 S   0.0   0.0   0:00.01 kthreadd
    3 root      20   0 0 0 0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-netns
   10 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
   13 root      0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0 0 0 0 S   0.0   0.0   0:00.29 ksoftirqd/0
   15 root      20   0 0 0 0 I   0.0   0.0   0:01.22 rcu_preempt
   16 root      20   0 0 0 0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
   17 root      20   0 0 0 0 S   0.0   0.0   0:00.04 rcu_exp_gp_kthread_worker
   18 root      rt   0 0 0 0 S   0.0   0.0   0:00.19 migration/0
   19 root     -51  0 0 0 0 S   0.0   0.0   0:00.00 idle_inject/0
   20 root      20   0 0 0 0 S   0.0   0.0   0:00.00 cpuhp/0
   21 root      20   0 0 0 0 S   0.0   0.0   0:00.00 cpuhp/1
```

Scince child terminated after 10 seconds

```
ubuntu@ubuntu:~$ ls /proc/2302
ls: cannot access '/proc/2302': No such file or directory
ubuntu@ubuntu:~$ nano practical3.c
ubuntu@ubuntu:~$ nano practical3.c
ubuntu@ubuntu:~$ nano practical3.c
ubuntu@ubuntu:~$ cat /proc/2302/status
cat: /proc/2302/status: No such file or directory
ubuntu@ubuntu:~$ ps -ef | grep practical3
ubuntu 7699 5772 0 08:32 pts/0 00:00:00 grep --color=auto practical3
```